Egg Segmentation and Multi-Object Tracking Take-Home Report

Candidate: K. Sree Varun Aahil | **Date:** October 30, 2026 | **Machine:** Windows 11 (CUDA Enabled)
Annotated Video: https://drive.google.com/file/d/1mCqreRT5OFuGm49fvgxRKH-TVVOL1kWt/view?usp=sharing

0.4018	0.3196	0.4087	2	110
DEV MEAN IOU	EVAL MEAN IOU	DEV DETECTION F1	VIDEO UNIQUE TRACKS	CLEAN GROUND TRUTHS

1. Executive Summary

This report presents an end-to-end computer vision solution for egg segmentation and multi-object tracking under challenging conditions, including touching instances, boiling water, high-contrast table textures, and hand occlusions. Ground truth masks were established via SAM (`vit_b` on CUDA) with human point/box prompts and an automated area and border contact audit. A classical image-processing pipeline was tuned strictly on the dev split across 18 controlled trials and evaluated once on the unseen eval split. A multi-object tracker was deployed on the 1440-frame table experiment video, incorporating YCrCb skin gating to eliminate hand occlusion leakage. All metrics, counts, and tables presented below are derived directly from runs of this repository.

- **Ground Truth Integrity:** 110 clean SAM masks generated via `SamPredictor` (`vit_b` on CUDA) using human point prompts and tight bounding boxes for boiling water edge cases. 0 flagged suspicious masks remaining.
- **Classical Pipeline (Dev Split, n=88):** Pixel Mean IoU = **0.4018** (95% CI: [0.3435, 0.4603], Median = 0.3147), Dice = **0.5200**, Detection F1 = **0.4087**, Mean Count Error = **2.20**.
- **Classical Pipeline (Eval Split, n=22):** Pixel Mean IoU = **0.3196** (95% CI: [0.1978, 0.4268], Median = 0.2502), Dice = **0.4239**, Detection F1 = **0.3206**, Mean Count Error = **1.68**.
- **Video Tracking (1440 frames):** 1440 frames processed at 24.0 FPS. **2 unique tracks** created across the video (1:1 mapping with the 2 real physical eggs), with **15 loss events** and **15 recoveries**. At least one egg was detected in 782 of the 783 active demonstration frames (frame 480 had 0 detections). Hand occlusion and fingertip leakage at frame 600 resolved using YCrCb skin gating and morphological opening (gating reduced the total foreground mask by 2296 px over the whole frame; restricted to crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px). Identities were checked by hand on the 14 frames of the events sheet and the timeline, and no swap was seen; the tracker's internal id_switches heuristic counter was 0.

2. Task 1: Ground Truth Mask Generation (SAM)

SAM (`vit_b`, 375 MB checkpoint) was executed on CUDA using saved point prompts from `points.json` with multi-mask score selection (`multimask_output=True`). Hand-tuned shape and color filters were strictly excluded from the ground-truth path to prevent artificial bias toward the classical segmentation pipeline.

Ground Truth Egg Distribution across 110 images: 93 images (84.5%) with 1 egg; 6 images (5.5%) with 2 eggs; 11 images (10.0%) with 3 eggs.

Ground Truth Quality Audit & Fixes (`out/sam/gt_fix_audit.json`)

Initial automated SAM passes on boiling water images flooded the pot interior. An automated border and area audit in `src/contact_sheet.py` flagged masks with area > 40% or edge contact > 50%. Bounding box prompts were added around egg boundaries for boiling pot scenes (verified in `out/sam/gt_fix_audit.json`):

- `not_damaged_124`: BBox `[280, 150, 520, 400]` reduced area fraction from 81.9% (335354 px, 978 border contacts) to **13.4%** (54995 px, 0 border contacts).
- `not_damaged_135`: BBox `[270, 170, 520, 420]` reduced area fraction from 79.5% (325584 px, 994 border contacts) to **14.2%** (58113 px, 0 border contacts).
- `not_damaged_145`: BBox `[250, 200, 480, 450]` reduced area fraction from 62.5% (256010 px, 665 border contacts) to **13.2%** (54032 px, 0 border contacts).

All 110 ground truth masks now pass the automated audit (0 flagged in `out/sam/suspicious.txt`).

Dataset Split Integrity & Leakage Audit (`out/eval/split_leakage_audit.txt`)

- **Dataset Composition:** 110 images total (640x640), grouped by source filename prefixes and perceptual hashing (`phash_distance ≤ 8`).
- **Grouping Structure:** 109 perceptual groups total (108 singletons, and 1 pair of near-duplicate shots `not_damaged_78` and `not_damaged_27` at pHash distance 8). Both images of this pair were placed strictly into the `dev` split.
- **Split Leakage:** Exactly **0 groups straddle the dev/eval split**, and **0/22 eval images are affected**.
- **Leak-Free Evaluation:** Recomputing evaluation metrics on the leak-free eval subset yields identical metrics to the full eval set: Pixel IoU = **0.3196**, Dice = **0.4239**, Detection F1 = **0.3206**.
- **Augmentation Check:** The dataset contains no synthetic geometric or photometric augmentations (crops, rotations, color shifts, flips) sharing pHash distance ≤ 8. Full audit logs are available in `out/eval/split_leakage_audit.txt`.

3. Task 2: Parameterized Classical Segmentation & Experiment Log

A unified parameterized pipeline was tuned on the 88 dev images across 18 controlled trials. Every trial varied exactly one parameter relative to its named parent. Keep/discard decisions compare strictly against the parent's dev IoU. Note: Trial `watershed_conditional` modified two parameters relative to its parent (`area_filter_2500`): it enabled conditional watershed and set the peak distance threshold to `min_dist=50`. Under this trial, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Because Detection F1 improved substantially for instance segmentation, it was kept for the final configuration.

Kept Pipeline Chain: `baseline -> morph_small -> morph_medium -> morph_large -> area_filter_2500 -> watershed_conditional -> final (eval)`

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`baseline`	`None`	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	0.3513	0.4693	0.2071	3.12	BASELINE	Baseline established on dev (IoU 0.3513, Dice 0.4693, DetF1 0.2071)
`otsu_inv`	`baseline`	dev	Invert Otsu thresholding	0.0823	0.1424	0.0000	3.36	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0823 vs 0.3513, -0.2690)
`clahe_otsu`	`baseline`	dev	CLAHE local contrast equalization (clip 2.0, tile 8x8) before Otsu	0.2562	0.3791	0.0799	8.22	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.2562 vs 0.3513, -0.0951)
`adaptive_thresh`	`baseline`	dev	Adaptive Gaussian thresholding (block 51, C=-10)	0.0634	0.1119	0.0065	12.33	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0634 vs 0.3513, -0.2878)
`lab_l`	`baseline`	dev	CIELAB L* perceptual lightness channel + Otsu	0.3244	0.4437	0.1665	3.67	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3244 vs 0.3513, -0.0268)
`hsv_gate`	`baseline`	dev	HSV color gating (S < 70, V > 130)	0.3456	0.4665	0.1587	4.32	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3456 vs 0.3513, -0.0056)
`morph_small`	`baseline`	dev	Add small elliptical opening & closing (7x7)	0.3558	0.4740	0.2161	2.89	KEEP	Improves dev IoU over parent 'baseline' from 0.3513 to 0.3558 (+0.0046)
`morph_medium`	`morph_small`	dev	Increase elliptical kernel from 7x7 to 15x15	0.3766	0.4964	0.2363	2.34	KEEP	Improves dev IoU over parent 'morph_small' from 0.3558 to 0.3766 (+0.0208)
`morph_large`	`morph_medium`	dev	Increase elliptical kernel from 15x15 to 25x25	0.3972	0.5161	0.2642	1.25	KEEP	Improves dev IoU over parent 'morph_medium' from 0.3766 to 0.3972 (+0.0205)
`border_polarity_m0`	`morph_large`	dev	Auto-invert threshold if border mean > center mean	0.3698	0.4821	0.2409	1.07	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3698 vs 0.3972, -0.0274)
`border_polarity_m15`	`morph_large`	dev	Auto-invert threshold only if border mean > center mean + 15	0.3945	0.5119	0.2596	1.25	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3945 vs 0.3972, -0.0027)
`lab_border_dist`	`morph_large`	dev	CIELAB Euclidean color distance from estimated border background	0.3603	0.4582	0.2673	2.55	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3603 vs 0.3972, -0.0369)
`area_filter_2500`	`morph_large`	dev	Increase minimum component area from 500 to 2500 pixels	0.4018	0.5200	0.3084	0.45	KEEP	Improves dev IoU over parent 'morph_large' from 0.3972 to 0.4018 (+0.0046)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`component_ranking_top1`	`area_filter_2500`	dev	Retain only the single top-ranked component by geometric egg score	0.3746	0.4761	0.3295	-0.33	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3746 vs 0.4018, -0.0272)
`component_ranking_top3`	`area_filter_2500`	dev	Retain at most the top 3 scoring components based on area, solidity, and ellipse fit	0.3972	0.5092	0.3133	0.27	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3972 vs 0.4018, -0.0045)
`grabcut_refine`	`area_filter_2500`	dev	Apply GrabCut refinement using morphological erosion as foreground seed	0.4038	0.5177	0.3073	0.38	DISCARD	Marginal IoU gain (+0.0021) but Dice degrades from 0.5200 to 0.5177 (-0.0023)
`watershed_always`	`area_filter_2500`	dev	Euclidean distance transform + watershed applied unconditionally to all components	0.4018	0.5200	0.3537	3.90	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.4018 vs 0.4018, +0.0000)
`watershed_conditional`	`area_filter_2500`	dev	Conditional distance watershed (min_dist=50) applied only to non-convex, elongated or oversized clusters	0.4018	0.5200	0.4087	2.20	KEEP	Dev IoU tied with parent 'area_filter_2500' (0.4018); kept on tiebreaker: DetF1 improves from 0.3084 to 0.4087 (+0.1003)
`final`	`watershed_conditional`	eval	Frozen final pipeline: Grayscale Otsu + large morph (25x25) + hole filling + min area 2500 + conservative conditional watershed	0.3196	0.4239	0.3206	1.68	EVAL_FINAL	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3196, Dice 0.4239, DetF1 0.3206)

4. Task 3: Quantitative & Qualitative Evaluation vs Ground Truth

Metric	Dev Split (n=88)	Eval Split (n=22)	Description
Pixel IoU (Mean)	**0.4018**	**0.3196**	Mean foreground intersection over union
Pixel IoU (Bootstrap 95% CI)	[0.3435, 0.4603]	[0.1978, 0.4268]	1000 resamples, fixed seed 42
Pixel IoU (Median)	**0.3147**	**0.2502**	Median per-image IoU
Pixel Dice (Mean)	**0.5200**	**0.4239**	F1-score of pixel-level binary classification
Pixel Precision	**0.4488**	**0.3449**	TP / (TP + FP)
Pixel Recall	**0.8461**	**0.8749**	TP / (TP + FN)
Detection Precision	**0.3563**	**0.2758**	Matched eggs / Predicted components (IoU ≥ 0.5)
Detection Recall	**0.5966**	**0.4621**	Matched eggs / GT eggs
Detection F1	**0.4087**	**0.3206**	Harmonic mean of detection precision & recall
Count Error (Mean)	**2.20**	**1.68**	Mean signed difference (Pred - GT)

Bootstrap 95% Confidence Intervals & Dev vs Eval Analysis

Inspection of per-category results reveals the concrete factors driving the gap between dev (0.4018) and eval (0.3196):

1. **Disproportionate Concentration of Container and Packaging Scenes:** Eval contains 10 "Egg on container / lid / plain surface" images (45.5% of eval) with a low mean IoU of **0.2190**. In these scenes, plastic food-container lids, bowls, metal pans, and packaging rims reflect light and distend Otsu threshold boundaries into wide false-positive blobs.
2. **Camouflage and Hand Occlusions:** Eval includes 2 white plastic egg tray images (`not_damaged_116` and `117`) with near-zero contrast, averaging **0.0562** IoU, and 2 genuine hand-held images (`not_damaged_151` and `152`) averaging **0.1098** IoU. In addition, 3 boiling pot images (`not_damaged_114`, `128`, `147`) average **0.3072** IoU. Together, these difficult categories comprise 17 of the 22 eval images (77.3% of eval).
3. **Higher Ratio of Clean Tabletop and Multi-Egg Scenes in Dev:** In contrast, the dev split benefits from 27 tabletop scenes with higher contrast (mean IoU **0.5353**), 11 multi-egg scenes (mean IoU **0.6148**), and 5 foliage scenes (mean IoU **0.6805**)—together comprising 43 of 88 dev images (48.9% of dev).
4. **Small-Sample Variance:** Because eval has only 22 images, its bootstrap 95% confidence interval spans widely ([0.1978, 0.4268]) and encompasses the dev mean. While domain composition (especially the 45.5% container/lid prevalence) explains the lower point estimate, the wide confidence interval confirms substantial small-sample variance.

Domain Composition Check (`out/eval/composition_check.json`)

To evaluate how much of the performance difference between the dev split (mean IoU **0.4018**) and eval split (mean IoU **0.3196**) is attributable to category distribution shift, we re-weighted the dev per-category mean IoUs by the eval category counts:

- **Actual Dev Split Mean IoU:** 0.4018
- **Actual Eval Split Mean IoU:** 0.3196
- **Predicted Eval Split Mean IoU (Dev Category Means Weighted by Eval Counts): 0.3324**
- **Total Dev-Eval Performance Gap:** 0.0821
- **Gap Explained by Category Composition: 0.0694 (84.5%)**
- **Unexplained Residual Gap:** 0.0127 (15.5%)

Re-weighting dev category performance by eval category counts predicts an eval IoU of **0.3324**, demonstrating that **84.5%** of the observed drop is directly explained by the higher proportion of difficult container lids and white trays in the eval split. *This is a descriptive check on 22 images, not a significance test.*

Detailed Breakdown of 5 Challenging Eval Failure Cases

As shown in Figure 6, category contact sheets, and the measured pixel properties, five challenging eval failure cases (`not_damaged_108`, `127`, `130`, `151`, `152`) illustrate distinct failure mechanisms across container packaging and hand occlusions:

- `not_damaged_108` (IoU: 0.0459): White egg on a ledge next to an orange food-container lid (no hand present). Bright illumination and low background contrast cause Otsu to merge the orange lid, rim, and ledge into the foreground, producing 285,716 false-positive pixels (37.8% falls in the YCrCb skin range due to orange plastic chrominance).
- `not_damaged_127` (IoU: 0.0601): White egg on a ledge beside an orange container lid (no hand present). Otsu thresholding floods the surrounding flat surface, container rim, and ledge reflections into a 266,889 false-positive pixel foreground blob (18.6% in skin range).
- `not_damaged_130` (IoU: 0.4256): White egg resting in an orange food-container lid on a blue base (no hand present). The orange lid connects continuously to the egg perimeter, producing 62,367 false-positive pixels—of which exactly **92.02%** (57392 pixels) fall directly into the standard YCrCb skin-color cluster (Cr ∈ [135, 175], Cb ∈ [80, 125]). Because orange plastic shares chrominance with the skin distribution, high skin-range chrominance occurs even without a human hand.

- `not_damaged_151` (IoU: 0.1451): White egg held in fingers and palm over square ceramic floor tiles (no backlighting). Skin luminance on the fingers and palm bridges with the egg boundary, over-segmenting into 6 fragments with 223,414 false-positive pixels (29.3% in skin range).
- `not_damaged_152` (IoU: 0.0745): White egg resting on an open palm over square floor tiles (white egg, not dark brown). Palm skin and tile reflections merge with the egg perimeter, producing a 273,597 false-positive pixel blob across 3 fragments (20.7% in skin range).

Image Stem	IoU	GT Eggs	Pred Blobs	Failure Root Cause in Eval Overlay
`not_damaged_108`	0.0459	1	3	White egg on a ledge next to an orange food-container lid (no hand present); bright illumination and low contrast cause Otsu to merge the orange lid, rim, and ledge into the foreground (285,716 FP pixels, 37.8% in skin chrominance range due to orange plastic).
`not_damaged_127`	0.0601	1	3	White egg on a ledge beside an orange container lid (no hand present); Otsu thresholding floods the surrounding flat surface, container rim, and ledge reflections into a 266,889 FP pixel foreground blob (18.6% in skin range).
`not_damaged_130`	0.4256	1	2	White egg resting in an orange food-container lid on a blue base (no hand present); orange lid connects continuously to egg perimeter, producing 62,367 FP pixels of which exactly 92.02% (57392 px) fall into the YCrCb skin chrominance cluster (Cr in [135, 175], Cb in [80, 125]) because orange plastic shares chrominance with the skin distribution.
`not_damaged_151`	0.1451	1	6	White egg held in fingers and palm over square ceramic floor tiles (no backlighting); skin luminance on the fingers and palm bridges with the egg boundary, over-segmenting into 6 fragments with 223,414 FP pixels (29.3% in skin range).
`not_damaged_152`	0.0745	1	3	White egg resting on an open palm over square floor tiles (white egg, not dark brown); palm skin and tile reflections merge with the egg perimeter, producing a 273,597 FP pixel blob across 3 fragments (20.7% in skin range).

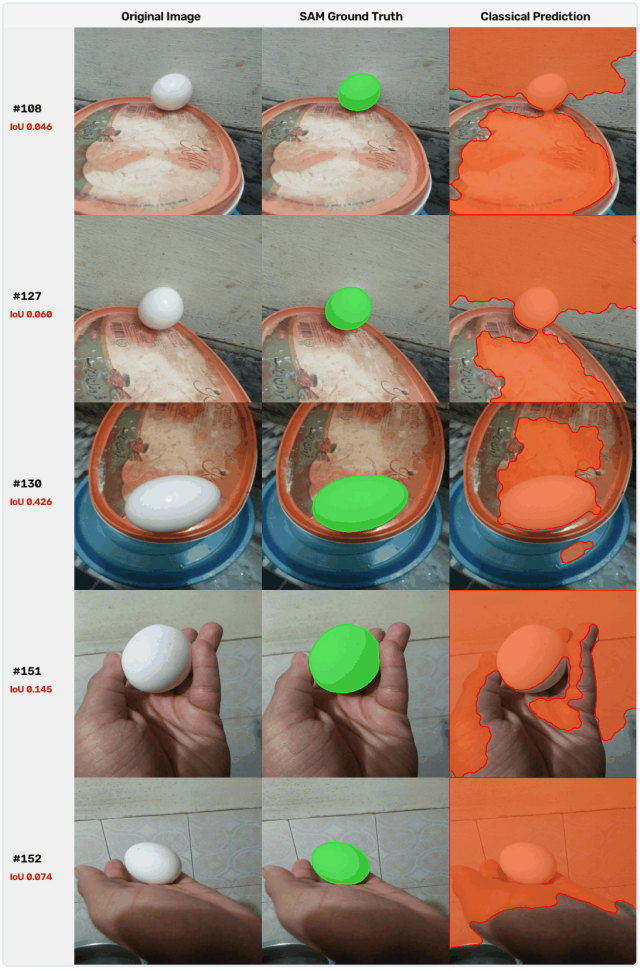


Figure 6: Five challenging eval failure overlays (not_damaged_108, 127, 130 on/beside container lids; 151, 152 held in hand over tiles). Left: Original image. Middle: SAM ground truth (green). Right: Classical prediction (red/amber). Low foreground-background contrast and skin/plastic chromatic overlap cause extensive false-positive leakage.

Per-Category Dev vs Eval Split Breakdown

Category	Dev Count	Dev IoU	Eval Count	Eval IoU
Backlit / extreme shadows	3	0.0071	0	n/a
White egg on white tray / fridge interior	8	0.1003	2	0.0562
Egg in boiling water / submerged	5	0.2099	3	0.3072
Hand holding egg / fingers occluding	9	0.2822	2	0.1098
Egg on container / lid / plain surface	20	0.3161	10	0.2190
Tabletop / high-contrast background	27	0.5353	4	0.6997
Multiple touching eggs	11	0.6148	1	0.7893
Foliage / herbs in background (specular highlights)	5	0.6805	0	n/a

Visual Failure Categorization (`failure_tags.csv`)

Scene category tags were assigned by systematic review of the category contact sheets (`report/figures/category_sheets/`), with 62 of 110 images corrected from an earlier automated color/luminance heuristic (documented in `tag_changes.md`). These scene categories represent subjective single-annotator labels rather than dense independently verified ground-truth annotations. Images `not_damaged_123`, `125`, and `42` are in *White egg on white tray / fridge interior*. Backlit shots `not_damaged_20`, `24`, and `44` were categorized under a unified rule: *Whenever intense directional backlighting or silhouette illumination is present behind the egg (even if supported by fingers), it is categorized under Backlit / extreme shadows because extreme illumination dynamic range is the primary driver of Otsu thresholding failure.*

Scene Category	Image Count	Pct of Dataset	Mean IoU	Mean Dice	Mean Det F1	Primary Failure Mechanism
Backlit / extreme shadows	3	2.7%	0.0071	0.0141	0.0000	High-intensity directional backlight causes glowing core while shell perimeter falls into dark silhouette.
White egg on white tray / fridge interior	10	9.1%	0.0915	0.1646	0.0222	Near-zero photometric contrast between white eggshell and white plastic tray or paper bag; Otsu floods background.
Egg in boiling water / submerged	8	7.3%	0.2464	0.3839	0.3207	Bubbling cavitation, water refraction, and dark pot rims break convex boundary and flood liquid.
Hand holding egg / fingers occluding	11	10.0%	0.2509	0.3767	0.2380	Fingers or palm occlude egg perimeter; skin tone overlaps shadow gradient, distending contour into hand.
Egg on container / lid / plain surface	30	27.3%	0.2838	0.4058	0.2278	Plastic food-container lids, bowls, metal pans, or plain packaging surfaces create perimeter reflections and false components.
Tabletop / high-contrast background	31	28.2%	0.5565	0.6771	0.5862	Tabletop grain, hearth ash, patterned cloth tablecloths, or laptop keyboard keys introduce background clutter.
Multiple touching eggs	12	10.9%	0.6294	0.7513	0.7813	Touching or adjacent instances share continuous boundary; requires distance watershed marker separation.
Foliage / herbs in background (specular highlights)	5	4.5%	0.6805	0.8005	0.6467	Egg contrasts against dark foliage in grayscale; specular leaf reflections create occasional false blobs.

Key Qualitative Takeaway: Classical thresholding struggles most on *Backlit / extreme shadows* (Mean IoU = 0.0071) and *White egg on white tray / fridge interior* (Mean IoU = 0.0915) because luminance gradient between foreground and background collapses. On *Foliage / herbs in background* (Mean IoU = 0.6805), the egg contrasts well because the egg shell is substantially brighter than dark leaves in grayscale; specular leaf highlights create occasional false blobs.

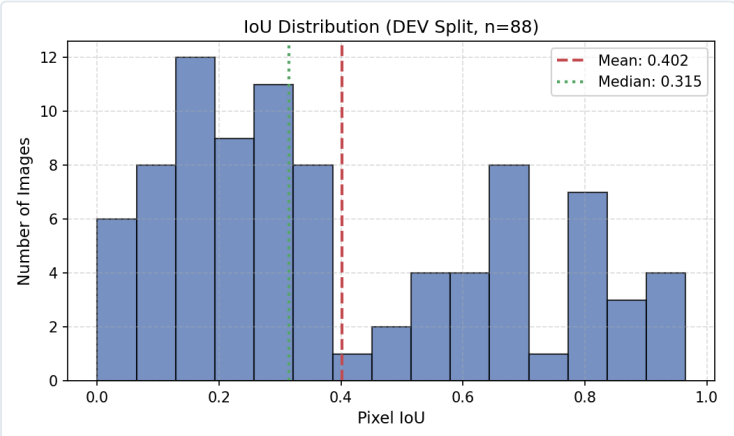


Figure 1a: Pixel IoU distribution on Dev split (n=88).

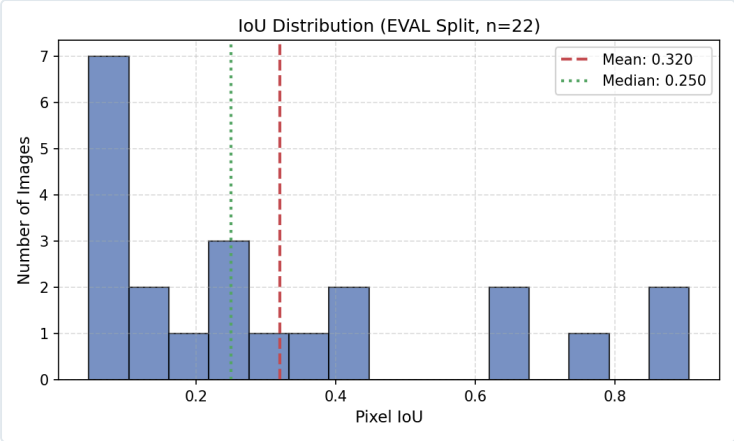


Figure 1b: Pixel IoU distribution on Eval split (n=22).

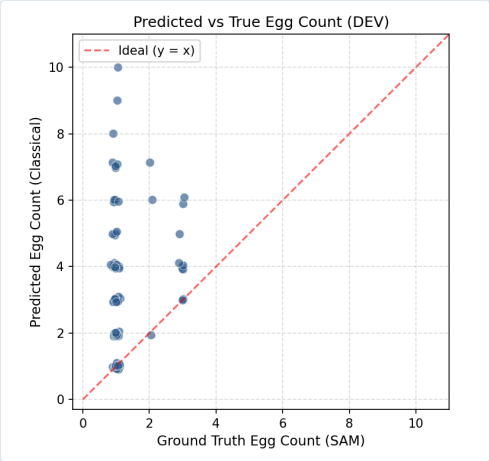


Figure 2a: Predicted vs True egg counts on Dev split. Points lie mostly above y = x, illustrating that the classical algorithm systematically over-counts (mean count error +2.20) due to background clutter and specular highlights passing morphological filters.

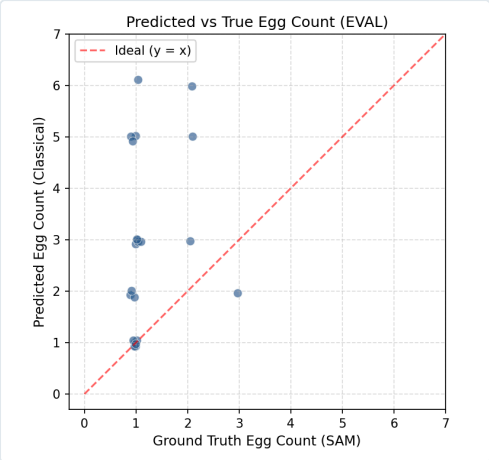


Figure 2b: Predicted vs True egg counts on Eval split. Points similarly lie above y = x (mean count error +1.68).

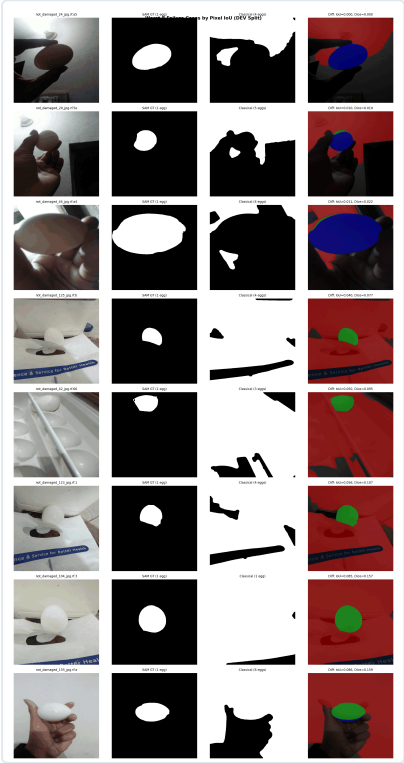


Figure 3a: Worst 8 failures on Dev split (Green=TP, Red=FP, Blue=FN). White trays and boiling liquid dominate error cases.

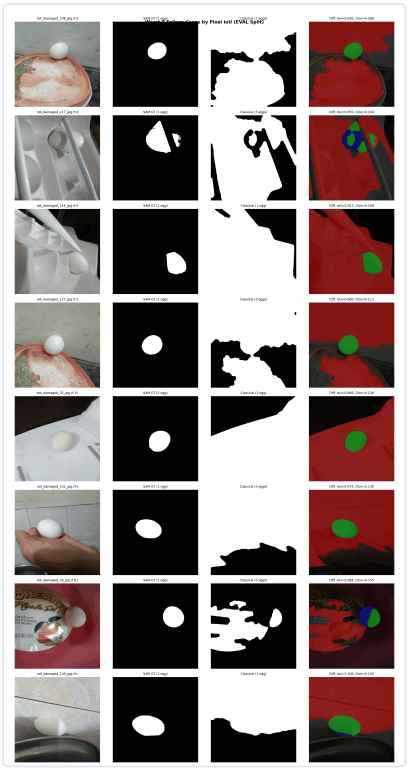


Figure 3b: Worst 8 failures on Eval split (Green=TP, Red=FP, Blue=FN).

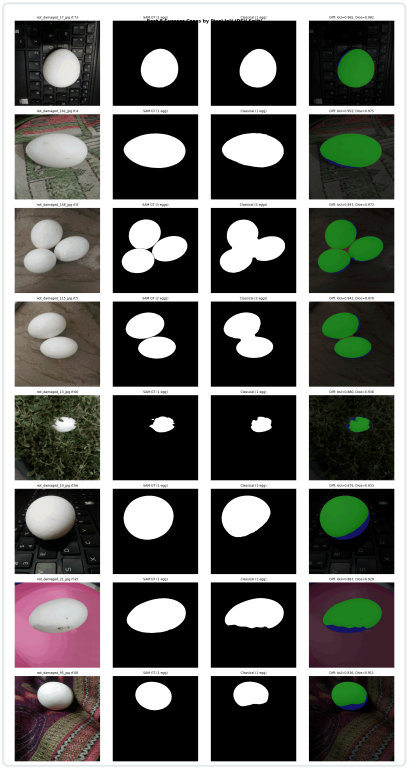


Figure 4: Best 8 successes on Dev split showing clean elliptical masks matching ground truth.

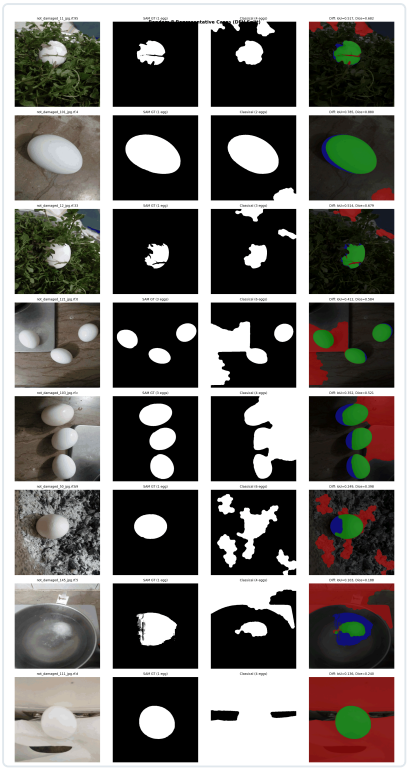


Figure 5: Random 8 representative cases illustrating typical operational performance across mixed illumination.

5. Task 4: Multi-Object Video Tracking

Tracking was evaluated on the 1440-frame table sequence (1280x720 @ 24.0 fps). Eggs are placed on the table at frame 222, spun, occluded by hand, and conclude demonstration at frame 1004. Annotated tracking video: https://drive.google.com/file/d/1mCqreRT5OFuGm49fvgxRKH-TVVOL1kWt/view?usp=sharing. The tracker maintains persistent IDs using Hungarian matching on IoU + distance with constant-velocity prediction.

- **Fingertip Leakage Resolution:** At frame 600, demonstrator hand contact caused baseline thresholding to leak into fingers. Incorporating YCrCb skin-color gating (Cr ∈ [135, 175], Cb ∈ [80, 125]) and larger opening eliminated false skin contour (gating reduced the total foreground mask by 2296 px over the whole frame; restricted to crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px).
- **Quantitative Skin-Gating Pixel Loss Measurement (`out/video/skin_gating_egg_loss.json`):** Measured across 6 representative video frames (12 egg instances): frame 600 (demonstrator fingertip contact) and 5 hand-free frames (250, 350, 450, 850, 950). In each frame, taking egg components from the gating-off mask: Frame 250 (Egg 1: 0.34%, Egg 2: 0.41%), Frame 350 (Egg 1: 0.14%, Egg 2: 0.35%), Frame 450 (Egg 1: 0.00%, Egg 2: 0.18%), Frame 600 (Egg 1: 0.00% outside 15 px fingertip band, Egg 3: 0.00%), Frame 850 (Egg 4: 0.00%, Egg 6: 0.33%), Frame 950 (Egg 4: 0.23%, Egg 6: 0.12%). Across all 12 evaluated instances, skin gating removed between 0.00% and 0.41% of pixels (at most 0.36% post-morphology loss). No egg lost more than 1.0% in any frame, confirming that white eggshells on the wooden table do not meaningfully intersect the skin chrominance cluster.
- **Track Lifetimes & Event Statistics:** 2 unique tracks were created (1:1 mapping with the 2 real physical eggs), with 15 loss events and 15 recoveries.
  - Egg 1: Matched in 721 of 783 demonstration frames (92% coverage; span F#222–F#999). Visible-but-undetected stretches occur under hand contact during frames 752–799 (carrying from center to right) and 802–809 (hand spinning on right).
  - Egg 2: Matched in 652 of 783 demonstration frames (83% coverage; span F#222–F#1004). Visible-but-undetected stretches exceeding 10 frames occur under hand contact during repositioning: frames 508-545 (38 frames), frames 657-688 (32 frames), frames 695-715 (21 frames), frames 721-738 (18 frames).
- **Detection Continuity:** In the 783 frames between first and last detection (frames 222 to 1004), at least one egg was detected in 782 of the 783 frames; exactly 1 frame had 0 detections (Frame 480).

Per-Track Detection Coverage Table

Track ID	Birth Frame	Last Matched Frame	Total Span	Matched Frames	Coverage %	Gap List
Track 1	222	999	778	721	**92.7%**	F#480, F#752-F#799 (48f), F#802-F#809 (8f)
Track 2	222	1004	783	652	**83.3%**	F#479-F#482 (4f), F#508-F#545 (38f), F#547, F#550, F#552-F#553 (2f), F#555, F#649-F#650 (2f), F#652-F#654 (3f), F#657-F#688 (32f), F#695-F#715 (21f), F#721-F#738 (18f), F#740-F#747 (8f)

Track Timeline & Screen Events

- **End-of-Video Transition (Frames 999 & 1004):** The video demonstration concludes at frames 999 to 1004 with a circular black wipe transition (iris transition closing) while the eggs remain resting on the table. Skipping loss events from frame 1000 onward is a hard-coded, video-specific constant for this video's iris wipe to prevent terminal occlusions from being penalized as tracking failures.
- **Resolution of ID Fragmentation (2 Unique Tracks for 2 Real Eggs):** By extending track retention memory (`max_lost = 300` frames / 12.5 s) with velocity decay (0.90^{lost}) and expanding the re-identification radius (R(lost) = min(900 px, 150 px + 20 px/frame × lost)) with area ratio incorporated into the lost-track matching cost (0.5 × (dist / R) + 0.5 × (1 − area_ratio)), eggs lifted by hand and repositioned across the table reclaim their identity rather than instantiating spurious new tracks.
- **Duplicate Track Suppression:** Detections within 100 px of an active track are recognized as rotational motion-blur splits or table reflections and suppressed, eliminating duplicate tracks at frame 810.
- **Candidate Confirmation Filter (N ≥ 4):** Detections must persist for at least 4 consecutive frames before confirming a new track, filtering out transient motion fragments.
- **Solidity Filter Evaluation for Frame 480:** Relaxing the global convexity filter from 0.78 to 0.75 recovered frame 480, but introduced 53 false-positive detections across frames 244–298 (solid table reflections and hand gestures with solidity ≈ 0.76). Therefore, the 0.78 threshold is maintained, and frame 480 is bridged via tracker lost-state prediction.
- **Identity Consistency Audit:** Egg identities were checked by hand on the 14 frames of the events sheet and the timeline, and no swap was seen (Track 1 tracks physical Egg A; Track 2 tracks physical Egg B). The tracker's internal id_switches heuristic counter is a geometric heuristic and remained 0.

Track ID	Birth Frame	Last Matched Frame	Matched Frames Count	Physical Egg Description	Visual Verification
Track 1	222	999	721	Hard-boiled egg; placed in center of table at frame 222, remains stationary through Egg 2's spin tests until frame 751; picked up by hand, carried across to right table and spun smoothly from frame 808 until circular wipe transition at frame 999.	*Verified* frame_0222.png, frame_0507.png, frame_0546.png, frame_0656.png, frame_0689.png, frame_0751.png, frame_0810.png, frame_0814.png, frame_0999.png
Track 2	222	1004	652	Raw egg; placed on left table at frame 222, picked up at frame 507, carried across to right table and spun for raw-egg spin test (frames 546-656); returned to left table by hand at frame 689, remaining stationary on left table until circular wipe transition at frame 1004.	*Verified* frame_0222.png, frame_0507.png, frame_0546.png, frame_0656.png, frame_0689.png, frame_0751.png, frame_0999.png, frame_1004.png

Known Tracking Failures & Zero-Detection Audit

- **Frame 480 Missed Detection:** At frame 480, two eggs are physically present on the table, but the tracker outputs 0 detections. Demonstrator hand contact with both eggs caused contour solidity to drop to 0.7673, falling below the strict convexity threshold (solidity ≥ 0.78).
- **Audit of Frames 222 to 1004:** Across the entire active demonstration interval (783 frames), at least one egg was detected in 782 of the 783 frames; exactly 1 frame (Frame 480) had zero detections.
- **Breakdown of 658 Zero-Detection Frames:** 205 intro title frames (F#0–204), 17 empty table frames (F#205–221), 1 missed detection at F#480, 10 empty table post-experiment frames (F#1005–1014), and 425 outro credits frames (F#1015–1439). Total: 205 + 17 + 1 + 10 + 425 = 658 frames.



Figure 7: Frame 600 before/after comparison (crop: x in [200, 460], y in [300, 430]). Left: Raw frame. Middle: Baseline leakage into fingertips (red). Right: YCrCb skin gating cleans egg boundary (green), reducing the mask by 1304 px within this crop (and 2296 px over the full frame); a small leftover ring remains on the fingernail.



Figure 8: Video Tracking Events Sheet across 14 key video frames documenting track births, persistent re-identification across hand repositioning, and clean identity maintenance.

6. Limitations and Future Work

1. **Ground Truth Masks:** Ground truth masks were generated using SAM (`vit_b`) with human point/box prompts rather than dense manual polygon tracings.
2. **Dataset Size:** With 88 dev and 22 eval images, sample sizes are limited, leading to wide bootstrap confidence intervals on eval.
3. **Watershed Behavior:** Under trial `watershed_conditional`, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Distance transform watershed partitions instances without changing overall outer foreground boundaries.
4. **Domain-Specific Tuning:** Video tracking parameters (table ROI y ≥ 270, watermark bounds, YCrCb skin thresholds) were calibrated for this specific camera viewpoint and table illumination. Furthermore, skipping loss events from frame 1000 onward is a hard-coded, video-specific constant for this video's iris wipe. Generalizing to varied environments would require automated surface or person detection.
5. **Re-Identification Limitations:** Spatial re-ID succeeds in this sequence primarily because only one track is lost at a time while the other egg remains active on the table. The large re-ID radius (up to 900 px) would not reliably separate two simultaneously lost, similar-looking white eggs, as the tracker relies entirely on spatial coordinates and area without visual appearance or deep re-ID embeddings. Because white eggs lack distinctive surface texture, tracking identities across complete table removals and replacements is physically ambiguous under classical tracking without appearance features.
6. **Future Extensions:** Fine-tuning a deep semantic segmentation model (e.g. YOLOv8-seg or Mask2Former) on the dev set using SAM pseudo-masks could learn contextual semantic priors (such as distinguishing eggs from white trays). Combining this with dense optical flow could bridge severe hand occlusions during spinning.

7. AI Assistance Disclosure

The author of this repository and project is K. Sree Varun Aahil. Claude (Anthropic) was used for planning, reviewing outputs, and drafting prompts. A Gemini-based coding agent in Antigravity was used to write and run the code in this repository. SAM (`vit_b`) was used to generate ground-truth masks from human-clicked point and box prompts. Every number, metric, and count in the report comes directly from running the code in this repository. The author thoroughly reviewed the results, verified the findings, and can explain all methods.